

Measuring and accounting for war destruction using human-reported and satellite-derived conflict data: Ukraine 2022-2023

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Abstract: Kenneth Hewitt's 1983 article detailed the destruction and death resulting from World War II aerial bombardments with a focus on German and Japanese cities. Given data limitations, his maps and analysis were not disaggregated below the city scale. We build on his pioneering efforts and the increasing availability of high-resolution conflict data to examine how our ability to measure and account for war destruction has changed in the last forty years, especially in narrowing the time-lag between conflict events and subsequent analyses. Our analysis of the impact of the war in Ukraine in the twenty months after the Russian invasion of February 2022 relies on three existing datasets; a Ukraine-wide time series of urban damage data derived from open Sentinel-1 radar satellite imagery as well as two geolocated and time-stamped conflict event datasets (ACLED and VIINA). By comparing the conflict events to the urban damage in a space-time analytical framework, we present a multivariate approach to document the dynamics of urban destruction and account for factors contributing to shared or divergent assessments of the place, timing, and intensity of war. These methods represent a step towards the harmonization of multi-method war impact assessments, especially in the face of disputed facts in the 'fog of war'.

Key Words: War impact assessment, synthetic aperture radar (SAR) coherent change detection, spatio-temporal analysis.

Academic research and media accounts of conflict—its causes, dynamics, and outcomes—have changed significantly since the publication of Kenneth Hewitt's (1983) provocative and poignant article in the *Annals of the Association of American Geographers* (AAG). Hewitt came to the study of conflict from his background in natural hazards research, especially in high mountain environments (e.g. Hewitt 1997). His 1983 article is devoid of reference to emerging contemporaneous trends in the discipline of geography on spatial analysis of conflict events, new territorialities including separatist movements consequent on the end of the colonial era, and heightened US-Soviet geopolitical tensions during the Second Cold War (O'Loughlin 2004). Apart from rare instances of activism-motivated works like Yves Lacoste's (1976) *La Géographie ça sert d'abord à faire la guerre* (*Geography is primarily used for making war*), the discipline had to wait until the 1980s to see sustained works on international and even later, civil wars. But like Lacoste and as exhorted by Richard Morrill in his AAG Presidential address (1984), Hewitt's article nonetheless helped re-invigorate geographic scholarly interest in contemporary political and conflict developments.

Like Hewitt, we appreciate the limitations of a focus on physical destruction in the study of war. "Crude as such measures are, assessments of area destruction offer a preliminary indicator of the physical basis of place annihilation" (Hewitt 1983, 264). But motivated by the hope that conflict-affected populations will engage in "the tenacious if pathetic resumption of life amid the rubble and 'killing ground' of their cities by bombed-out survivors seems more fully to measure our topic and its tragedy than all the statistics of what was lost" (Hewitt 1983, 278), we exploit the availability of nationwide, satellite time series data for urban damage analysis, and combine it with media-derived conflict event data. These datasets offer much more systematic, timely, and consequential damage assessments compared to damage maps produced through aerial reconnaissance or field-based methods to which Hewitt referred. Still, damage maps remain incomplete or unavailable for conflicts in remote regions. In this paper, we conduct an exploratory analysis of war dynamics in Ukraine after Russia's full-scale invasion in February 2022 to examine the ways in which these new approaches document the place, timing, and intensity of war destruction of the urban physical environment in similar or divergent ways.

Contemporary research on conflict dynamics and assessing war damage

Hewitt's article was a clarion call for research on the devastating outcomes of modern wars on civilians, focused on the massive air raids on German and Japanese cities in the last years of World War II. 'Place annihilation' in the form of destruction of buildings (with more than 50 percent of German cities and one-quarter of Japanese urban housing stock razed, according to Hewitt) was not just the annihilation of physical structures but also of place meaning, histories, identities and belonging. Hewitt's data were aggregated to the city level using archival and published sources since detailed timestamped, georeferenced conflict event data were unavailable. For the past quarter-century, disaggregated information on war dynamics (such as Armed Conflict Location and Event Data (ACLED) and the Violent Incident Information from News Articles (VIINA) data for Ukraine that we examine in this paper) has revolutionized conflict research and provided near real-time information on diverse conflict impacts. More locationally and thematically accurate data are likely collected by various militaries but these remain generally inaccessible for academic analysis and public discussion except through leaks

(O'Loughlin et al. 2010). Wartime fieldwork and detailed mapping in bombed places, as demonstrated effectively over fifty years ago by Lacoste (1973) to highlight the impacts of U.S. bombings of North Vietnam's dike infrastructure, are both dangerous and rare so the ability of militaries to hide the effects of their actions, especially in relatively inaccessible regions, is paramount. However, the recent advent of free and openly-accessible Earth observation imagery combined with appropriate technical analysis is shifting the balance away from secrecy to greater public knowledge and resultant political actions (Witmer 2015; Mueller et al. 2021; Sticher, Wegner, and Pfeifle 2023; Scher and Van Den Hoek 2025).

Merging spatial analytical tools with georeferenced conflict event data has allowed insights into the geographic diffusion of violence (O'Loughlin and Witmer 2011) and the tit-for-tat dynamics commonly seen in wars (Linke, Witmer, and O'Loughlin 2012). In a parallel and related research field since the early 1990s about the causes and onset of wars, a renewed interest in the classic political geographic question of territories has been evident. While territorial control retains its 'allure' (Murphy 2013), territoriality—the management and control of space—is a state strategy that can be turned on and off (Paasi and Ferdoush 2022). In a world of sovereignty and borders, states claim four rights: to territorial jurisdiction, to non-intervention, to control their borders, and to regulate the use and extraction of resources in their territories (Stilz 2019). Each of these strategies may manifest in different spatio-temporal patterns of violence, within or beyond a state's borders.

International conflicts are now much less frequent than civil wars but two invasions by large states against smaller neighbors since Hewitt's 1983 article (Iraq against Kuwait in 1990 and Russia against Ukraine in 2014 and a full-scale invasion in 2022) call "into question the very right of one of the units making up the system to occupy a discrete space as a titular nation-state positioned next to a powerful neighbor on the world political map" (Murphy 2022). As of early 2026, Vladimir Putin's continued insistence on a peace treaty that includes both territorial changes (four Ukrainian oblasts and Crimea to be recognized as Russian territory) and an arrangement with Western powers that prevents Ukraine from joining NATO illustrates the close alignment of the geopolitical and the territorial elements of modern interstate conflict.

Modern war damage is often devastating across multiple domains—from visible destruction of buildings, infrastructure such as roads, railroads and phone/electricity networks to the effects of war trauma on people's psychology and wellbeing. In one of the best documented urban conflicts, 35-40% of homes in Bosnia-Herzegovina were destroyed in fighting or demolished in concerted efforts at 'domicide' designed to erase traces of previous residents by newly victorious armies (Toal and Dahlman 2011). A new field of forensic architecture research integrates evidence from multiple sources such as witness testimony, remote sensing, crowd-sourcing, and social and mass media to investigate building destruction and human rights abuses resulting from armed conflicts (Weizman 2017).

War damage in Ukraine

Targeting of key infrastructures has been a feature of the Russian missile attacks in Ukraine, both military and key civilian energy components. The effect has been to exaggerate a pre-2022

trend of a declining bureaucratic functioning of the Ukrainian state in establishing service provision and state legitimacy. The state bureaucratic capacity to deliver socioeconomic goods and services (e.g. healthcare, education, jobs, and infrastructure) has been severely compromised and thus, long-term post-war human development in Ukraine will be very slow to recover from an already low level (Clem, Herron, and Hoheneder 2024).

War damage extends beyond urban buildings and the legacy of bombing typically lasts much longer than the actual fighting in terms of abandoned and unusable agricultural land due to the presence of mines, unexploded ordnance, or other toxic war materials (Witmer and O'Loughlin 2009). Structural damage typically results in the loss of power and water supply, further amplifying risks to public health, government services and economic life. Nivievskyi and Neyter (2024) calculate that landmines are distributed across approximately 16 million hectares of agricultural land in Ukraine and that agricultural reparation alone will cost at least 3 billion USD and as much as 27 billion USD. As of December 2024, the total cost of reconstruction and recovery in Ukraine over the next decade is an estimated 524 billion USD (World Bank 2025).

The war damage to date has left profound impacts on civilian lives and welfare. In February 2025, nearly four million Ukrainians were displaced inside the country due to the Russian invasion with another seven million outside, mostly in European countries. Fully 12.7 million people were in need of humanitarian assistance according to the UNHCR (<https://www.unrefugees.org/emergencies/ukraine/>). Casualty figures on both the civilian and military side are highly politicized and unreliable but there seems little doubt that as of 2025, they amount to tens of thousands of Ukrainian civilians and hundreds of thousands of deaths in the combined militaries of Russia and Ukraine (Ivshina 2025). In Ukraine, surveys showed that in the first year (2022) of the full-scale invasion of Ukraine, only 7.2% of samples of displaced and non-displaced civilians showed no symptoms of severe or mild stress, anxiety, and PTSD (post traumatic stress disorder). The PTSD diagnosis standard was met by 50.8% of non-displaced civilians, 55.4% of internally displaced persons and 62.2% of refugees (Lushchak et al. 2024). A more recent (summer 2024) assessment revealed that only 51% of Ukrainians can meet their own or dependents' nutritional needs, 31% have witnessed missile or drone attacks, and 20% have heard or seen fighting (reSCORE 2024).

The challenges of documenting conflict dynamics and their impacts are apparent in Ukraine and pervasive across armed conflict settings. Satellite-based assessments, the de facto standard approach for urban damage mapping, involves visual interpretation of commercial very-high resolution (VHR) images with a spatial resolution of 1 meter or less. Restrictions on access to VHR imagery, incomplete geographic coverage over conflict zones, and intermittent or otherwise short-term assessments limit their value for studying nationwide impacts of protracted conflicts like the ongoing war in Ukraine (Bennett et al. 2022). For example, the United Nations Satellite Centre (UNOSAT) has conducted damage analysis in Ukraine using this visual interpretation approach across twenty-five cities with typically one or two assessments for each city with the last assessment completed on 14 October 2022 (as of this writing). These conflict impact assessments by UNOSAT use before/after (bi-temporal) imagery, which can preclude short-term or ephemeral changes (such as conflict-induced landscape degradation) that are better captured with long-term multi-temporal analyses (Van Den Hoek 2021).

Over the past quarter-century, fueled by the availability of internet sources including old (newspapers) and new (social) media, more disaggregated data on the time and location of conflict events have become readily available for mapping and analysis (Gleditsch et al. 2002). The advantages and deficits of the databases are debated but best practice suggests use of different sources to capture the full range, actors, victims and casualties of conflict events. While ACLED (Raleigh et al. 2010) and data from the Uppsala Conflict Data Program (UCDP; Sundberg and Melander 2013) have been extended to global coverage and provide near real-time information, other data sources are collated for specific types and locations of violence such as the machine-coded VIINA (Zhukov and Ayers 2023) data collected on the war in Ukraine.

In this paper, we examine the spatio-temporal patterns of wartime damage in Ukraine and consider the impact of a key difference between human-reported conflict event data and satellite image-driven assessments: a conflict event does not necessarily manifest in measurable landscape change but satellite image-based approaches are only able to capture events that result in measurable changes to the environment. To examine the degree and conditions of agreement between the two approaches, we use VIINA and ACLED georeferenced conflict event data and Sentinel-1 radar-derived damage data to evaluate the timing and geographic distributions of damage in Ukraine over the 20-month period of March 2022 through October 2023.

Measuring war destruction and dynamics

Conflict event data

We use georeferenced and time-stamped conflict event data from two datasets, VIINA (Zhukov and Ayers 2023) and ACLED (Raleigh et al. 2010). For events that occur in remote areas, the geolocated point is typically assigned to the nearest town or urban area. This loss of precision is one source of error in the datasets, though we attempt to minimize geolocation errors in this paper by omitting events whose geo-precision is insufficient for the spatial unit of analysis that we use, e.g. we remove province-level events. After filtering these less precise events and event types unlikely to result in structural damage, 60,012 VIINA events and 61,331 ACLED events remained (Table S1). See the Supplementary Material for additional details on the conflict event datasets.

Satellite-derived urban damage

Numerous Earth observation studies have characterized consequences of the full-scale Russian invasion on Ukraine's energy (UNDP and World Bank 2023) and healthcare infrastructures (Poole et al. 2025). We analyze the war damage data produced by Scher and Van Den Hoek (2025) who used medium-resolution synthetic aperture radar (SAR) data from the Sentinel-1 constellation and coherent change detection (CCD) techniques to map urban damage across Ukraine at a monthly time step from March 2022 through October 2023, approximating the first

year and a half of the conflict. (See Scher and Van Den Hoek (2025) for a detailed description of the SAR damage mapping methodology and Hernandez et al. (2024) for an interactive visualization of the data; this SAR technique was also applied to the recent US/Israeli bombing in Iran (Motevalli et al. 2026).) We note here aspects of the dataset pertinent for comparison with ACLED and VIINA data. First, the SAR-based damage analysis was restricted to urban areas that were structurally stable in the pre-war period; a requirement given that ‘damage’ was conceptualized as a sudden and sustained disturbance in urban SAR signal similarity relative to pre-war SAR signal characteristics. Second, the detection of damage must be confirmed for two consecutive months after the initial detection. Third, a minimum mapping unit of four connected pixels (6400 m²) was applied that excludes small, standalone damage in areas with low built-up density based on the Global Human Settlement (GHS) gridded data (Pesaresi et al. 2024). SAR damage data were temporally aggregated to produce a monthly, nationwide damage map at 40 meter spatial resolution. The SAR damage data do not measure repeat damage and only account for the first occurrence of pixel-level damage.

Visual analysis of spatio-temporal conflict data

Charting conflict timelines

For the time series plots, we collapse the spatial dimension and aggregate our datasets to show monthly changes over time across all of Ukraine. We count the number of events per month for the ACLED and VIINA datasets and measure the SAR-detected damage area as a percentage of the total urban area recorded by GHS. Despite the overall similarity in total number of media-derived events, the monthly counts show considerable divergence, especially in the initial months of the war (Figure 1). The ACLED counts remain below 1500 in each of the first two months while the VIINA data register two to three times as many events. The counts converge for much of the summer and fall of 2022 before diverging again for most of 2023. Figure 1 also shows the trend for SAR-derived damage with the greatest damage occurring in the first two months of the war followed by a secondary spike in late 2022 before settling to a lower level of damage in 2023.

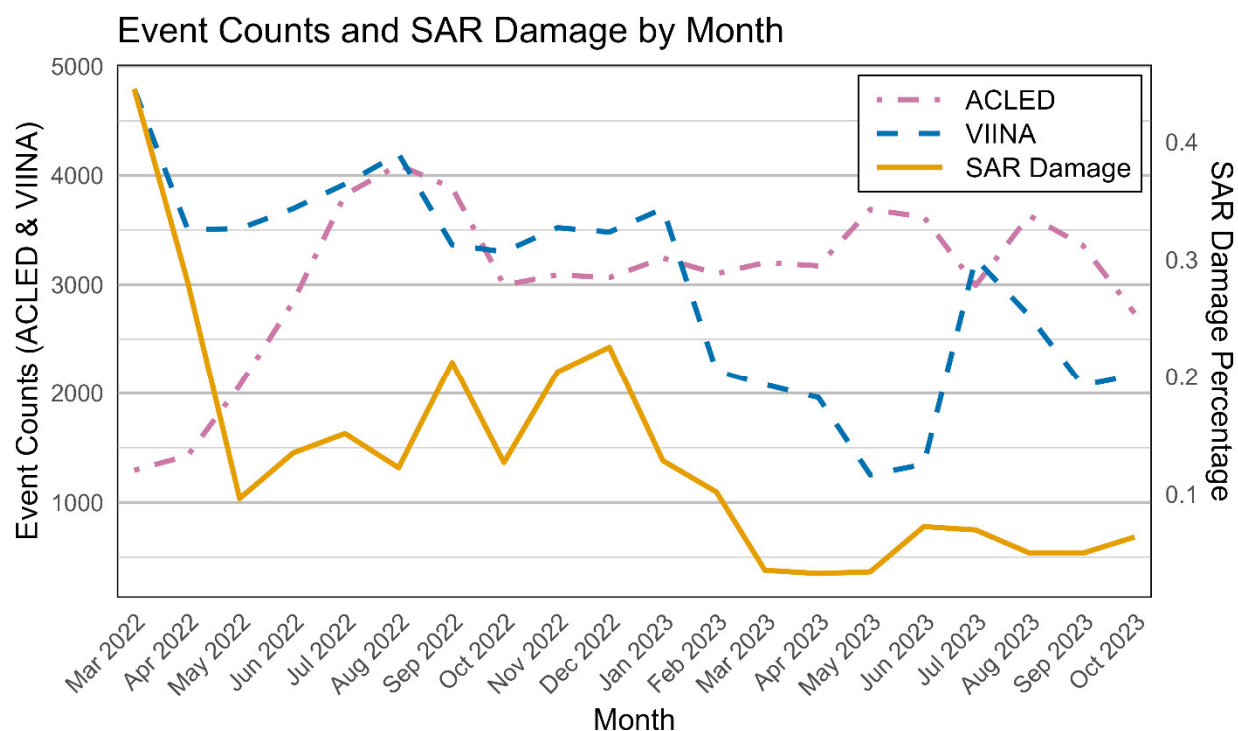


Figure 1. Time series plots of media-reported events and SAR-derived damage as a percentage of GHS urban area.

A visual comparison of the media-derived events and satellite imagery trends suggests that the SAR damage and VIINA event counts are more aligned in picking up the high intensity of conflict seen in the early months of the war. The SAR damage area declines dramatically after the first couple of months whereas both VIINA and ACLED remain relatively elevated. The decline in SAR damage is largely due to the inability to detect repeat damage at the same location, as mentioned above. This is especially noticeable in districts like Rubizhanska, Popasnianska, and Mariupolska which contributed to the high March and April 2022 SAR-detected damage signal, but whose signal drops off after that despite continued fighting after these three districts fell into the Russian-controlled zone.

Mapping the war

For the spatial data visualization, we aggregated event and damage data to the *hromada* administrative district level. Ukraine's administrative divisions include twenty-seven *oblasts* (provinces)/cities at level 1, 136 *rayons* (level two districts), and over 1400 *hromadas* (level three communities). We use geographic border data from the Humanitarian Data Exchange (OCHA 2024) to aggregate our data to a common spatial unit of analysis. For any study using geographic data, the choice of spatial unit can affect the conclusions (Linke et al. 2017). Cook and Weidmann (2022) discuss the spatial aggregation challenge as it relates specifically to event data. They suggest, among other things, a consideration of both the spatial scale of the political processes as well as the quality of the data when selecting a scale for analysis. Since our focus is on the community-level effects of war, we elected to use the *hromada* districts

(Figure 2) as these spatial units align with existing infrastructure and the built environment. This scale should also minimize some of the geo-coding error that occurs when trying to precisely locate individual events based on media reports.

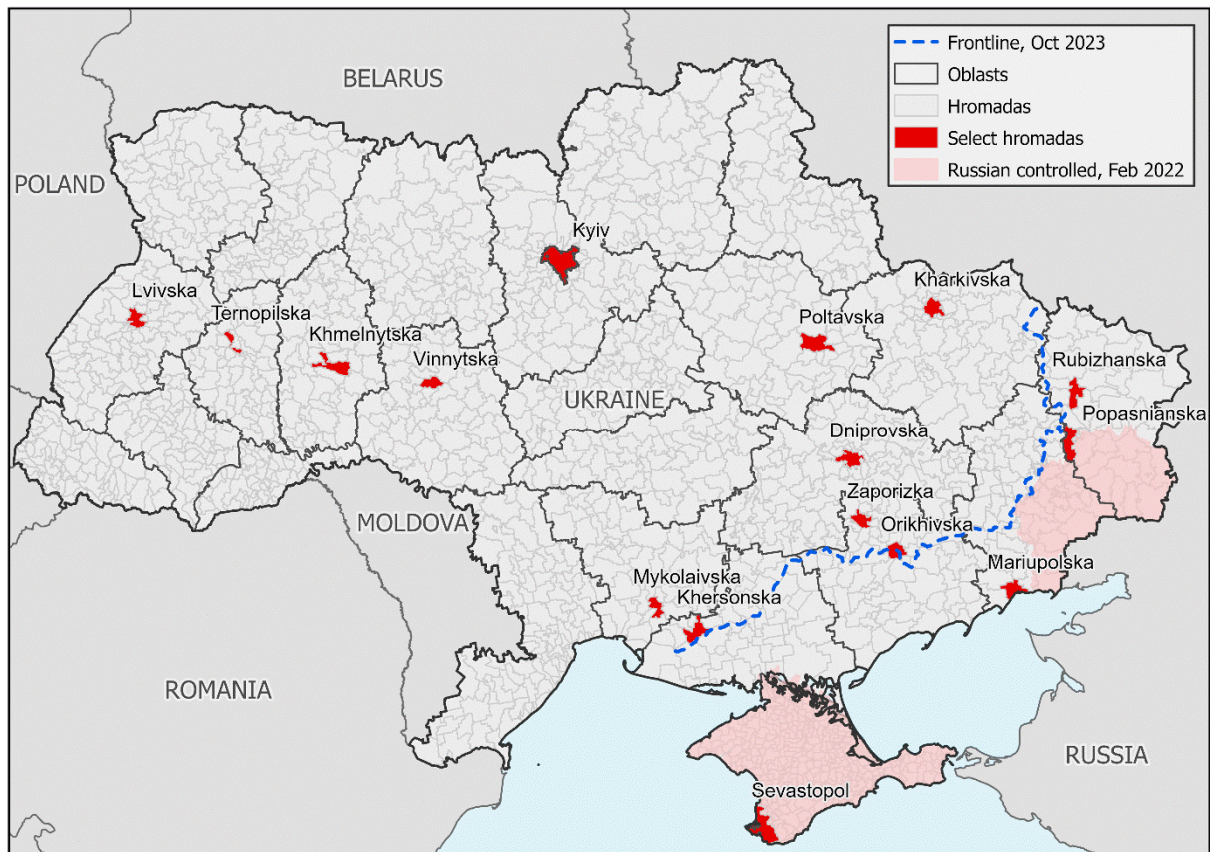


Figure 2. Overview map of Ukraine territory prior to 2014 with oblast boundaries (dark grey borders) and the level 3 *hromada* community administrative districts (light grey borders) with select districts mentioned in the text labeled and in red. The blue dashed line represents the frontline as of October 2023.

The first set of maps (Figures 3 and 4) shows the national distributions of each dataset in aggregate over the study period. For the event data, we normalized the event counts by dividing by the area (km²) of each *hromada*. To reduce the effect of outliers in the visual representation, we use a non-linear set of categorical breaks.

Though all three datasets show the most intense fighting along the frontlines, there are notable differences. For the media events, the VIINA map (Figure 3A) shows many more events in major cities farther away from the frontlines in comparison to the ACLED map (Figure 3B). This difference is especially noticeable for Kyiv, but also for other major urban centers such as Kharkiv, Dnipro, Zaporizhzhia, Kherson, Mykolaiv and Mariupol. Overall, VIINA has a broader geographic inclusion across the spectrum of rural-urban settings beyond the frontlines. In a few areas, ACLED records more events, such as the districts northeast of Kherson along the Dnipro River.

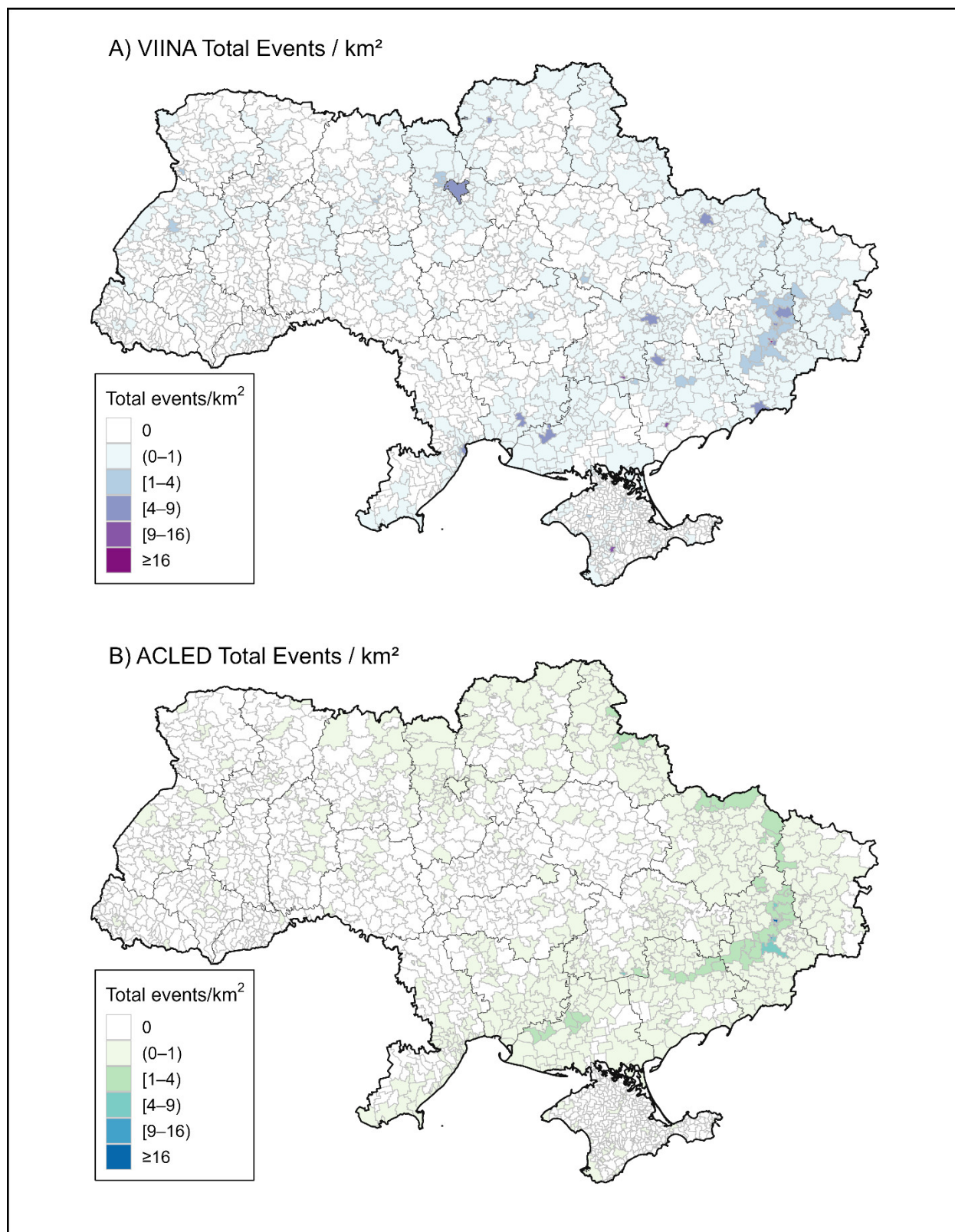


Figure 3. A) VIINA and B) ACLED event density for the study period March 2022 to October 2023.

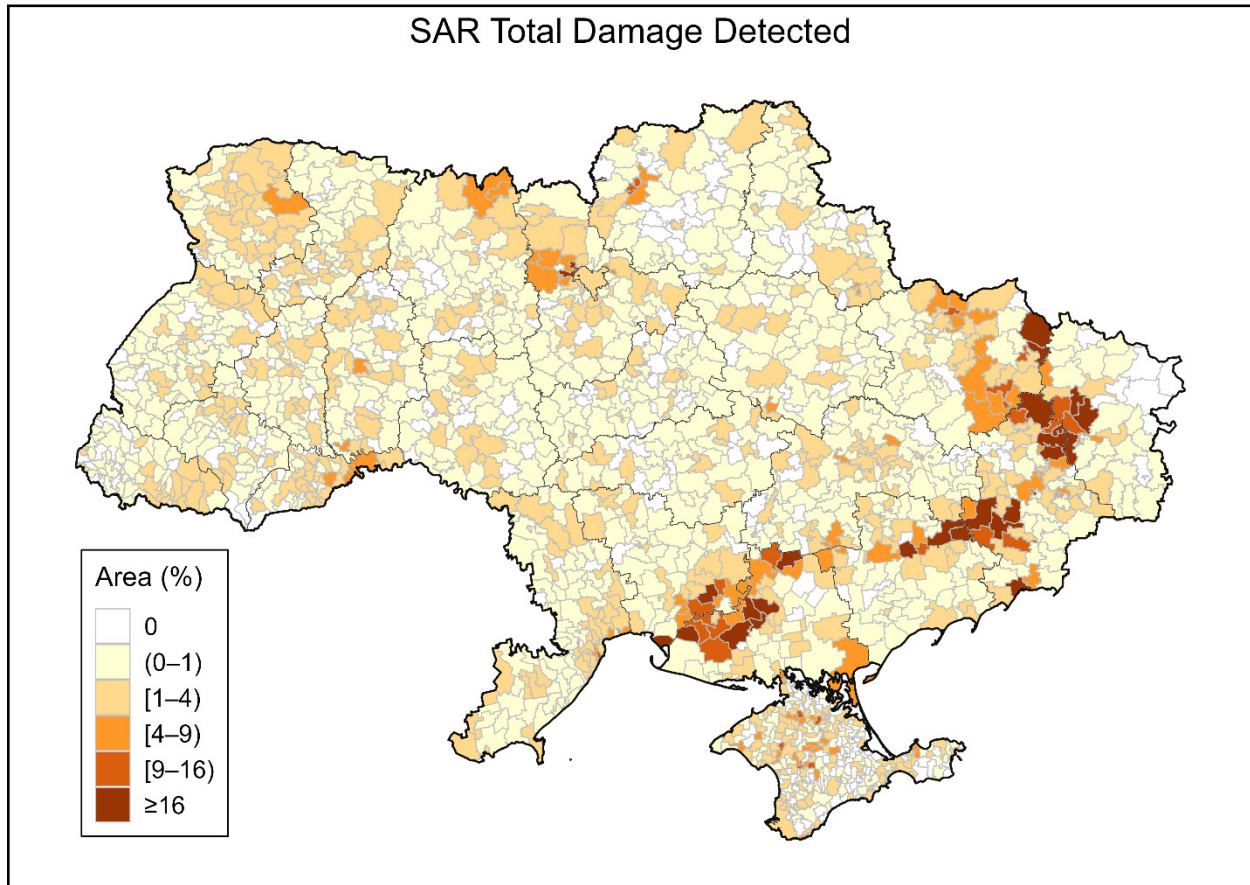


Figure 4. Total SAR damage (March 2022 to October 2023) as a percentage of GHS area.

For the SAR damage data, we calculated the area of pixels classified as damaged and we normalized this value by the total GHS urban area within each *hromada*. While this method of pixel counting has the potential to produce biased estimators, the lack of ground truth reference data precludes the calculation of unbiased area estimates (Olofsson et al. 2013). The map in Figure 4 shows the distribution of damage across much of Ukraine, with the frontlines and the city of Mariupol especially visible. The city of Orikhiv (northwest of Mariupol on the frontline) has been devastated by bombings, with an estimated 80% of buildings destroyed (Ptak 2024). Neither of the media event databases reflect this high level of destruction, likely because a nighttime barrage of missiles and shelling might result in one or two events in a database, but has the effect of damaging dozens or more structures.

Gauging complementarity and agreement of event and satellite data

To capture the persistence of violence over the 20-month study period, we create a categorical *hromada*-level map that shows the number of months with both a media event (either VIINA or

ACLED) and SAR-detected damage (Figure 5). This representation reveals some patterns not apparent in the preceding maps. Though conflict is consistently reported along the frontline, other districts with major urban centers in western Ukraine are also highlighted. In particular, L'viv, Ternopil, Khmelnytskyi, Vinnytsia, Poltava and Sevastopol experienced regular violence throughout the study period.

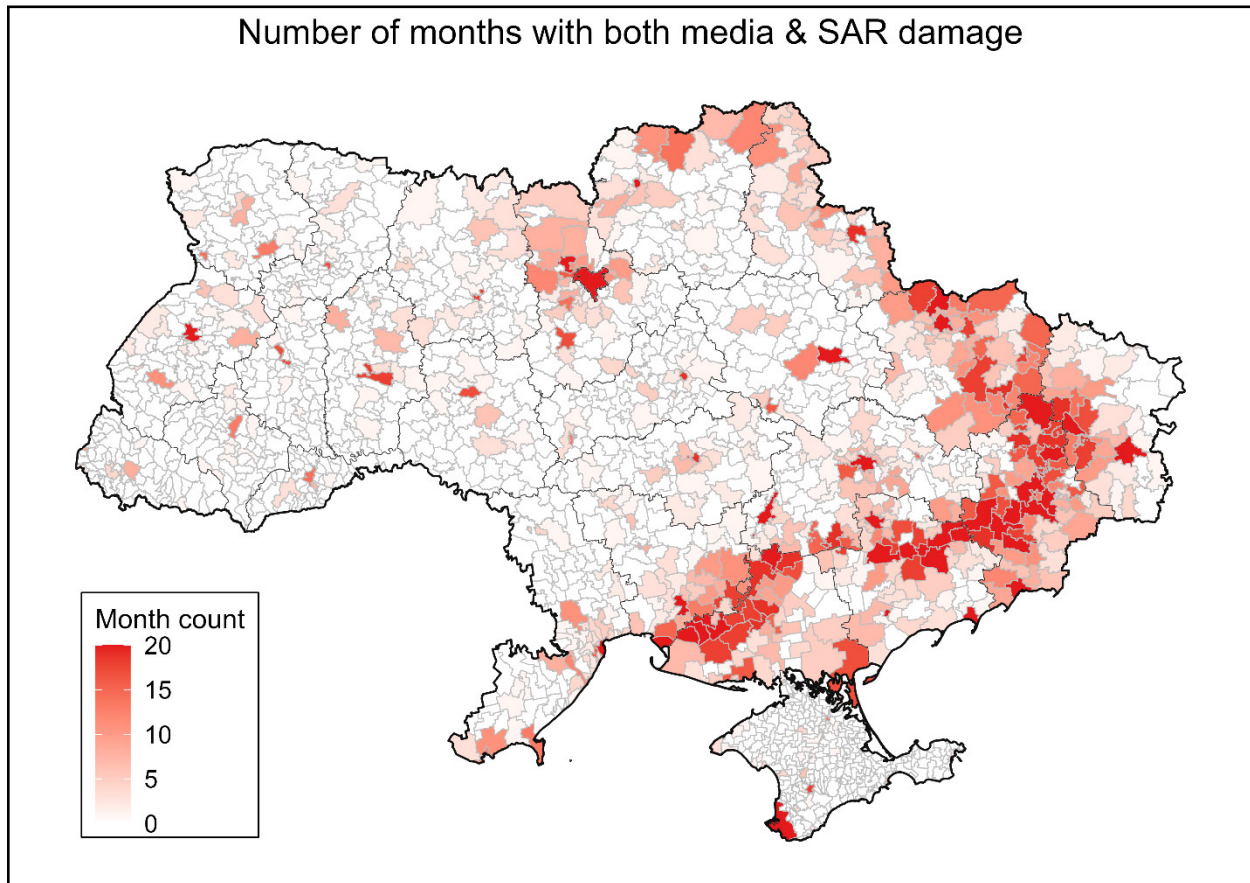


Figure 5. Categorical map showing the number of months with both a media event (VIINA or ACLED) and SAR damage.

We next compare the persistence of damage reports by counting the number of months when each *hromada* district has a media report separately for the VIINA and ACLED datasets and similarly count the number of months of SAR damage. We combine these counts in plots with the number of months of media events on the y-axis and SAR damage months on the x-axis (Figure 6). The size of each dot is scaled by the total SAR damage area for the district. If the media reports and SAR imagery are capturing the same events perfectly, we would expect a large dot for each month along the diagonal. Instead, we see in both plots that the SAR damage data are capturing damage in districts in months without corresponding media events. For districts with more than fifteen months of media events, large amounts of damage are also detected by the satellite imagery. For the VIINA data (Figure 6A), the total SAR area was 2.6 km², and 2.3 km² for the ACLED data (Figure 6B). Continuing to sum the SAR damage area into

5-month bins, the middle bins (6-10 and 11-15 months) each have 0.3 - 0.4 km², while the smallest bin (0-5 months) totals are 1.2 km² for VIINA and 1.4 km² for ACLED. This suggests that for areas experiencing the heaviest fighting, both the SAR data and media reports succeed in capturing the war dynamics, but for areas with little media coverage, there is a large amount of damage that remains unreported, with ACLED appearing to miss more SAR-detected damage than the VIINA database. Animations of the space-time dimensionality of the conflicts can be seen at this website: <https://github.com/fwitmer/ukraine-damage>.

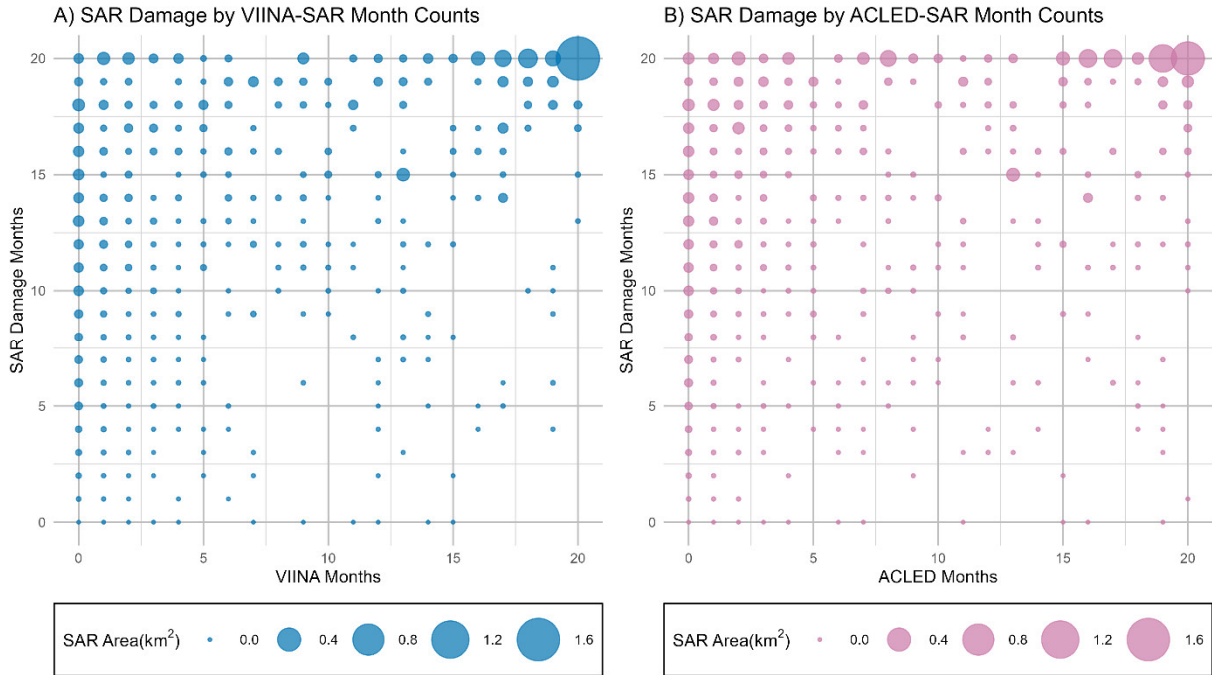


Figure 6. SAR damage area by the number of months with media reported events, A) VIINA and B) ACLED, and SAR-detected damage. The counts and area summations are by *hromada* district.

Correlation analysis of media and satellite coverage

In statistically comparing our datasets, we expect a degree of similarity because the data-generating events on the ground are the same, but that similarity does not always result in reaching the database inclusion threshold. VIINA and ACLED events must be witnessed and reported by humans while SAR has no requirement for human observation. The datasets are also differently sensitive to conflict impacts: an event must yield a measurable physical impact to be detected by SAR but VIINA and ACLED have no such requirement.

To calculate the correlation statistics, we convert the data from a ‘wide’ format to a ‘long’ format where each row represents a space-time value for the *hromada*-month. The values are the same area-normalized media counts and SAR damage area described above. Table 1 shows correlation coefficients for these three space-time variables. Since each variable is highly

skewed with many values equal or close to 0, we report the non-parametric Spearman's rank and Kendall's tau coefficients, in addition to the Pearson correlation coefficient. We also include spatio-temporal filtered versions of the correlation coefficient in the Supplementary Material as a robustness check. As expected, all coefficients are positive, though the strength of the association is lower than we anticipated, revealing the extent to which these datasets include conflict events differently. The VIINA-ACLED correlation is strongest with a Pearson correlation coefficient of 0.50, with events in one dataset only explaining about 25% of the variation in the other. The correlation between the SAR-detected damage and media events is weaker, with the Pearson correlation nearly the same at 0.20 for both media datasets, and the VIINA data a bit more correlated with SAR damage data for the Spearman and Kendall metrics. The versions of the correlation coefficients that control for spatial and temporal dependence show similar relationships between the three measures (Table S2) and effectively remove the spatial dependence as measured by Moran's *I* (Table S3).

Table 1. *Hromada*-month correlation coefficients.

	Pearson	Spearman	Kendall
VIINA-SAR	0.20	0.28	0.24
ACLED-SAR	0.20	0.25	0.22
VIINA-ACLED	0.50	0.56	0.53

Composite results - making a damage estimate

We use the results from the statistical analysis to inform our decisions for creating a composite damage score of Ukraine. We conclude that the media event datasets and SAR-derived damage estimates are largely capturing different events or aspects of events within the same conflict and thus, we elected to weight the media and satellite datasets equally in the creation of a composite map. The media event datasets, though different, are sufficiently similar to combine as input for half of the composite map. We use the following equation to calculate our composite damage score based on cumulative *hromada* area-normalized media counts and GHS-normalized SAR damage recorded over the study period:

$$\text{CompositeDamage} = \text{mean}(\text{VIINA}, \text{ACLED}) + \text{SAR damage}$$

To minimize the effect of outliers, we transform the inputs using a square root distribution scaled from 0 to 50. When added together, this yields a final composite damage score ranging from 0 to 100 where 0 indicates an absence of conflict and damage in each dataset while 100 indicates the highest recorded presence of conflict and/or damage across the combined media and satellite datasets. Figure 7 shows the resulting composite damage map. The distribution reflects the low-level of SAR-detected damage in many small urban areas throughout Ukraine as well as

heavy fighting and damage along the frontlines and persistent shelling and bombing of urban centers well beyond the frontline that is often well represented in all three datasets.

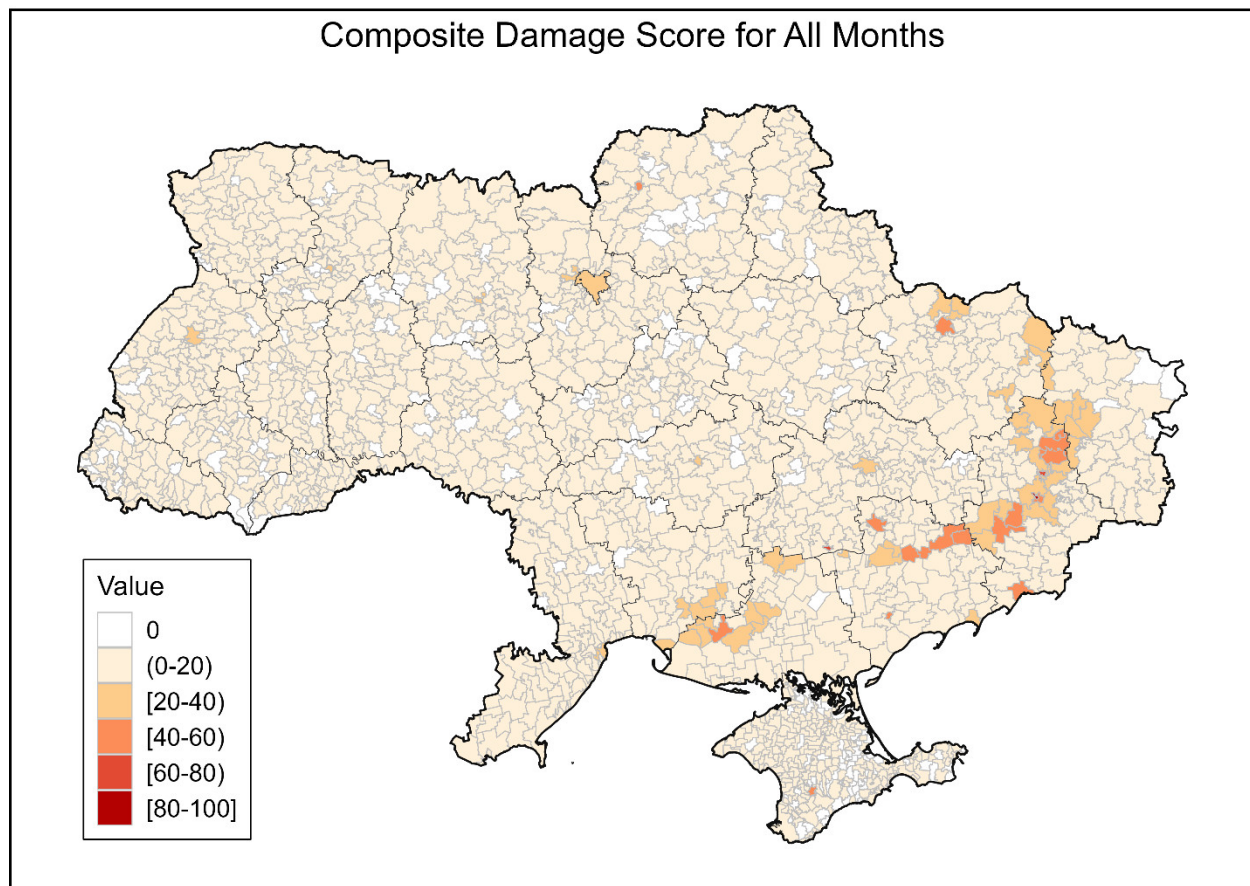


Figure 7. Composite damage score combining the media events with SAR-derived damage.

Conclusion: Improving the documentation of destruction

Using the most comprehensive publicly available conflict and damage datasets, we traced the patterns of violence in Ukraine for the first 20 months of the war after the Russian invasion in February 2022. The media-based reports in both ACLED and VIINA are incomplete since they typically fail to capture all conflict events in remote, dangerous or inaccessible places. In the case of Ukraine, drone or missile attacks on major centers far from the frontline are likely to appear in media databases but the many conflict events that constitute the fighting on the frontline are relatively undercounted. Attacks on civilian infrastructure and residential buildings resulting in civilian casualties are more likely to be reported than military ones.

The radar damage data from Scher and Van Den Hoek (2025) partially compensates for the missing information in the media reports, especially in remote regions where reporting on conflict event occurrence presents additional challenges. While not considered in this retrospective assessment, the rapid, automated generation of urban damage assessments

using open-access Earth observation data also offers a currency unmatched by media datasets, which require additional time for data collation and validation. Still, remote sensing of damage using satellite SAR data can have many limitations with regard to detection of low severity damage, repeat damage, and damage to small, isolated structures. Further, ground truth building-level damage data appropriate for validation of radar-derived damage data are broadly lacking, and overhead optical damage assessments like those conducted by the United Nations Satellite Centre (UNOSAT) are not appropriate for validating radar-based damage assessments due to differences in detection methodology and spatial resolution (Humanitarian OpenStreetMap Team 2025). Without such ground truth data, pixel-based estimates of damage cannot be validated or bias-corrected following best practices (Olofsson et al. 2013). Finally, satellite data do not include rich thematic information common to media reports such as conflict actors or the specific type of conflict event that resulted in physical damage, challenging the attribution of detected damage with specific conflict events.

The advent of geo-referenced conflict event and satellite-derived damage mapping enabled us to document destruction in Ukraine in ways not possible four decades ago when Hewitt (1983) published his article about destruction in Germany and Japan using historical archival information from nearly forty years earlier. While Ukraine is probably the world's most documented conflict due to the wide availability of crowd-sourced reports of violence across the country as well as the determination of the Kyiv government to report attacks far from the frontline, the 'fog of war' precludes full accounting for the scale and scope of all war events. The relative contribution of satellite-derived information and human coded events will inevitably differ between conflicts with current wars in Sudan and Ukraine demonstrating vast differences in both kinds of coverage. Our study has indicated the desirability of both kinds of documentation and suggests the careful consideration of data sources (and their omissions) for conflict analysis. Such a synoptic view of a war taking place over vast distances provides insights not possible from individual news articles or photographs, though many details are missing when looking at destruction through the lenses of media reports and satellite imagery. This study's explicit recognition of the complementary nature of media-reported and satellite-identified conflict events and damage is a promising avenue for future research. While significant technical advances will continue, the reduction of war to rows in a spreadsheet or pixels in an image fail to capture the immediate and visceral impact of conflict on the lives of the people caught up in it and the landscapes that bear witness.

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Measuring and accounting for war destruction using human-reported and satellite-derived conflict data: Ukraine 2022-2023

Supplementary Material

Conflict event data

Geolocating and time-stamping conflict events from media reports is always a challenge. For events that occur in remote areas, the geolocated point is typically assigned to the nearest town or urban area. This loss of precision is one source of error in the datasets, though we attempt to minimize geolocation errors in this paper by omitting events whose geo-precision is insufficient for the spatial unit of analysis that we use. We discuss this aggregation and choice of spatial units below for the VIINA and ACLED datasets. We opted not to use the Uppsala Conflict Data Program (UCDP) Georeferenced Event Dataset (GED) since it requires at least one death for an event to be included in the dataset. This higher threshold means there are many fewer events, many of which presumably resulted in damage to buildings and infrastructure. Furthermore, the UCDP data are coded in such a way that prevents filtering by conflict events likely to result in damage to buildings and infrastructure.

VIINA

The VIINA dataset started production two weeks into the current Russia-Ukraine war at the beginning of March 2022. These data are updated daily and available for download via a github repository (Zhukov and Ayers 2023). Instead of relying on international news sources, VIINA uses local news sources in the Ukrainian and Russian languages in addition to English sources and aggregators such as the Live Universal Awareness Map (Liveuamap, <https://liveuamap.com/>). About 92.5% of the sources are Ukrainian in origin, and 7.5% are Russian. Though a few research papers acknowledge the existence of this new VIINA dataset, only a few have published using it, either by overlaying the VIINA data onto their maps (Bartusevičius et al. 2023; Goto et al. 2023) or by conducting a textual analysis (Maia Polo, Saxena, and Alexopoulos 2024).

To code the content of each event, VIINA relies on a long short-term memory (LSTM) recurrent neural network to extract information such as actors, type of event, and geolocation. For our study, we are only interested in events likely to cause damage to buildings and infrastructure. We therefore subset the data by only retaining events for the following categories: air strike (includes strategic bombing and helicopter strike), tank battle, field artillery, improvised explosive device (includes land mine and car bomb), and destruction of property or infrastructure. We further subset the VIINA data based on the geo-precision associated with

each event. VIINA maintains a dictionary of Ukrainian place names mentioned in media and uses Yandex and OpenStreetMap to retrieve the geographic coordinates for each place in event reports and record the precision of each place (e.g., a specific administrative level). We filtered the data to only include municipal (“ADM3”) and “STREET” level data, with provincial (“ADM1”) and district (“ADM2”) level events removed. The “STREET” level data are typically georeferenced to a metro station or a city district. This geoprecision filter reduced the number of VIINA events from 77,928 to 60,012 for the study period (Table S1). Of these remaining events, only 3,180 (5.3%) have “STREET” level geoprecision. Our initial aim to conduct a local comparison of event data to satellite-derived estimates was prevented by this coarse level of geoprecision. ACLED data are similarly limited with best geoprecision at the town-level.

Table S1. Conflict events data filtering summary

	VIINA	ACLED
Total conflict events	77,928	61,816
After geoprecision filtering	60,012	61,331

ACLED

The ACLED dataset relies on human coding of media events associated with political conflict. The project began in the early 2000s by geocoding events from media for many countries in Africa (Raleigh et al. 2010). It expanded to global coverage in 2022, and has included Ukraine since 2018. As the more established dataset, a wider range of studies have been published using the Ukraine ACLED data than for VIINA.

We subset the ACLED event data to only include events likely to result in damage or destruction to infrastructure. The overarching event type for this is “Explosions/Remote violence”, which includes sub-event types of “Air/drone strike”, “Grenade”, “Remote explosive/landmine/IED”, “Shelling/artillery/missile attack”, and “Suicide bomb”. More than 88% of the events are in the shelling category with most of the remaining events associated with air/drone strikes.

We further subset the data to include only events with geographic precision at the level of town or small region; province-level events were excluded. This only reduced the number of events by 485, leaving us with 61,331 explosive-related events (Table S1), very similar to the 60,012 VIINA events for analysis.

Correlation analysis robustness checks

Since our data exhibit spatial and temporal autocorrelation, we use a spatial and temporal filtering method to remove this bias from the correlation coefficients reported in the main text. We do this by estimating autoregressive linear models for each variable and then calculating the correlation coefficient on the model residuals. This residual-based spatial filtering follows the

approach of Griffith & Peres-Neto (2006) which builds on the spatial lag concept introduced by Anselin (1988). We use three functional forms: 1) a spatial lag model, 2) a spatio-temporal lag model where the spatial lag is also lagged one month (the average neighboring violence in the prior month), and 3) a model with two separate autoregressive predictors, one spatial and one temporal. Table S2 reports the correlation coefficients of the model residuals. As expected, the relationships between each measure are lower than those reported in the main text after removing the spatial and temporal dependence inherent in the data. The key conclusion still holds, that the VIINA and ACLED data are much more similar than either's correlation with the SAR imagery.

Table S2. *Hromada*-month correlation coefficients after spatial and temporal filtering.

	Spatial lag	Spatio-temporal lag	Spatial plus temporal lag
VIINA-SAR	0.22	0.21	0.07
ACLED-SAR	0.18	0.19	0.06
VIINA-ACLED	0.33	0.35	0.20

To confirm that the filtering technique removed spatial dependence in the data, we calculated the Moran's *I* measure using the Queen's contiguity neighbors for each of the original variables and their corresponding autoregressive model residuals (Table S3). Each of the original variables exhibited strong spatial autocorrelation with mean monthly p-values < 0.001. For the model residuals, some of the monthly Moran's *I* values were still statistically significant, but on average, the filtering technique effectively removed the spatial autocorrelation from each measure.

Table S3. Mean monthly Moran's *I* p-values.

	Original	Spatial lag	Spatio-temporal lag	Spatial plus temporal lag
VIINA	$9.7 * 10^{-4}$	0.561	0.514	0.183
ACLED	$1.9 * 10^{-148}$	1.000	0.789	0.199
SAR	$1.4 * 10^{-20}$	1.000	0.299	0.647

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